Lecture 11 – Systematic Program Proofs

AAA551: Programming Language Theory

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PLRG

2026 Spring

Recall

- Axiomatic Semantics
  - Hoare Triples
  - Partial Correctness vs. Total Correctness
  - Assertion Language
  - Denotational Semantics of Assertions
  - Satisfaction
  - Validity
- Hoare Logic
  - Example – Factorial
  - Soundness and Completeness
  - Relative Completeness

Contents

1. Weakest Liberal Precondition
   - Motivation
   - Weakest Liberal Precondition
   - Example – Factorial
   - Provability and Correctness of $\mathbf{wlp}$

2. Dafny
   - Example 1 – Factorial
   - Example 2 – Fibonacci
   - Example 3 – IndexOf

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Motivation

Hoare logic is **non-deterministic** in its application, as it requires 'guessing' appropriate intermediate assertions for sequential composition and finding suitable assertions for the rule of consequence.

$$\textsc{Seq}\ \frac{\vdash \{\phi\}\ s_1\ \{\phi''\} \qquad \vdash \{\phi''\}\ s_2\ \{\phi'\}}{\vdash \{\phi\}\ s_1\,;s_2\ \{\phi'\}}$$

$$\textsc{Consequence}\ \frac{\vdash \phi \Rightarrow \phi_1 \qquad \vdash \{\phi_1\}\ s\ \{\phi_1'\} \qquad \vdash \phi_1' \Rightarrow \phi'}{\vdash \{\phi\}\ s\ \{\phi'\}}$$

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To resolve this non-determinism, we can use **the weakest liberal precondition**, which allows us to calculate the necessary assertions mechanically by reasoning backward from the postcondition.

Weakest Liberal Precondition

Intuitively, the weakest liberal precondition of a statement s with respect to a postcondition ϕ is the weakest assertion that guarantees that ϕ holds after executing s.

Definition (Weakest Liberal Precondition)

$\mathbf{wlp}(s, \phi)$ is the **weakest liberal precondition** of statement s with respect to postcondition ϕ if:

$$\forall \sigma, I.\ \sigma \models_I \mathbf{wlp}(s, \phi) \iff S[\![s]\!](\sigma) \models_I \phi$$

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Assertions $x < 3$ or $x < 5$ would both be sufficient preconditions for x := x + 1 to ensure that $x < 10$ holds afterwards.

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Assertions $x < 3$ or $x < 5$ would both be sufficient preconditions for x := x + 1 to ensure that $x < 10$ holds afterwards.

$$\mathbf{wlp}(x := x + 1, x < 10) = x < 9$$

However, the weakest liberal precondition is the least restrictive assertion that still guarantees the postcondition. In this case, it would be $x < 9$.

Weakest Liberal Precondition

$$\mathbf{wlp}(\texttt{skip}, \phi) \quad = \quad \phi$$

Weakest Liberal Precondition

$$\begin{aligned}
\mathbf{wlp}(\texttt{skip}, \phi) &= \phi \\
\mathbf{wlp}(x \texttt{ := } a, \phi) &= \phi[x \mapsto a]
\end{aligned}$$

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\mathbf{wlp}(\texttt{if } b \texttt{ then } s_1 \texttt{ else } s_2, \phi) &= (b \Rightarrow \mathbf{wlp}(s_1, \phi)) \wedge \\
&\quad\ (\neg b \Rightarrow \mathbf{wlp}(s_2, \phi))
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 & & (\neg b \Rightarrow \mathbf{wlp}(s_2, \phi)) \\
\mathbf{wlp}(\texttt{while } b \texttt{ do } s, \phi) & = & \mathbf{gfp} F = \bigwedge_{i \geq 0} F^i(\texttt{true})
\end{array}
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where

$$F(\phi') = (b \Rightarrow \mathbf{wlp}(s, \phi')) \wedge (\neg b \Rightarrow \phi)$$

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where

$$F(\phi') = (b \Rightarrow \mathbf{wlp}(s, \phi')) \wedge (\neg b \Rightarrow \phi)$$

However, it is impossible to compute $\mathbf{wlp}$ for loops in general, as it may require an infinite number of iterations to reach a fixed point.

Weakest Liberal Precondition – Loop Invariant

To overcome the challenge of computing $\mathbf{wlp}$ for loops, we can introduce the concept of a **loop invariant**, which is an assertion that holds before and after each iteration of the loop.

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Assume that a **loop invariant** $\phi_{\mathbb{I}}$ is given for $\texttt{while}\ b\ \texttt{do}\ s$:

$$\texttt{while}\ b\ @[\phi_{\mathbb{I}}]\ \texttt{do}\ s$$

Then, we can modify the definition of $\mathbf{wlp}$ for loops as follows:

$$\mathbf{wlp}(\texttt{while}\ b\ @[\phi_{\mathbb{I}}]\ \texttt{do}\ s, \phi) = (b \wedge \phi_{\mathbb{I}} \Rightarrow \mathbf{wlp}(s, \phi_{\mathbb{I}})) \wedge (\neg b \wedge \phi_{\mathbb{I}} \Rightarrow \phi)$$

Example – Factorial

```
while (0 < i) @[{0 ≤ n ∧ 0 ≤ i ≤ n ∧ x × i! = n!}] {

    x = x * i;

    i = i - 1;

}
```

$\{x = n!\}$

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Example – Factorial

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while (0 < i) @[{0 ≤ n ∧ 0 ≤ i ≤ n ∧ x × i! = n!}] {
    x = x * i;
    {0 ≤ n ∧ 0 < i ≤ n + 1 ∧ x × (i − 1)! = n!}
    i = i - 1;
    {0 ≤ n ∧ 0 ≤ i ≤ n ∧ x × i! = n!}
}
{x = n!}
```

Example – Factorial

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while (0 < i) @[{0 ≤ n ∧ 0 ≤ i ≤ n ∧ x × i! = n!}] {
    {0 ≤ n ∧ 0 < i ≤ n + 1 ∧ x × i! = n!}
    x = x * i;
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    i = i - 1;
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{x = n!}
```

Example – Factorial

$\{0 \leq n \wedge 0 \leq i \leq n \wedge x \times i! = n!\}$

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while (0 < i) @[{0 ≤ n ∧ 0 ≤ i ≤ n ∧ x × i! = n!}] {
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    i = i - 1;
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```
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```

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The precondition implies the $\mathbf{wlp}$ of the statement with postcondition:

$$\{0 \leq n \wedge i = n \wedge x = 1\} \implies \{0 \leq n \wedge 0 \leq i \leq n \wedge x \times i! = n!\}$$

Thus, we can conclude that the Hoare triple is valid:

$$\vdash \{0 \leq n \wedge i = n \wedge x = 1\}\ s\ \{x = n!\}$$

Provability and Correctness of $\mathbf{wlp}$

Theorem (Provability of $\mathbf{wlp}$)

For any statement s and assertion ϕ, the following holds:

$$\vdash \{\mathbf{wlp}(s, \phi)\}\ s\ \{\phi\}$$

The Hoare triple formed by $\mathbf{wlp}$ is not only semantically valid but also formally derivable ($\vdash$) using the deductive rules of Hoare logic.

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Theorem (Correctness of $\mathbf{wlp}$)

For any statement s and assertions ϕ and ϕ', the following holds:

$$\models \{\mathbf{wlp}(s, \phi')\}\ s\ \{\phi'\} \quad \wedge \quad \models \{\phi\}\ s\ \{\phi'\} \quad \Longrightarrow \quad \models \phi \Rightarrow \mathbf{wlp}(s, \phi')$$

Since $\mathbf{wlp}$ is the weakest assertion that guarantees the postcondition, any valid Hoare triple with the same statement and postcondition must have a precondition that implies $\mathbf{wlp}$.

Relative Completeness of Hoare Logic

Recall the definition of relative completeness:

Theorem (Relative Completeness)

If we assume that all valid assertions are provable in the assertion logic, then all valid partial correctness statements are derivable in Hoare logic:

$$(\forall \phi.\ \models \phi \implies \vdash \phi) \implies (\forall \phi, s, \phi'.\ \models \{\phi\}\ s\ \{\phi'\} \implies \vdash \{\phi\}\ s\ \{\phi'\})$$

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We can prove this by using the $\mathbf{wlp}$.

- By the correctness of $\mathbf{wlp}$, we have: $\models \phi \Rightarrow \mathbf{wlp}(s, \phi')$.

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- By the correctness of $\mathbf{wlp}$, we have: $\models \phi \Rightarrow \mathbf{wlp}(s, \phi')$.
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- By the provability of $\mathbf{wlp}$, we have: $\vdash \{\mathbf{wlp}(s, \phi')\}\ s\ \{\phi'\}$.

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- By the assumption of the oracle (all valid assertions are provable), we can derive: $\vdash \phi \Rightarrow \mathbf{wlp}(s, \phi')$.
- By the provability of $\mathbf{wlp}$, we have: $\vdash \{\mathbf{wlp}(s, \phi')\}\ s\ \{\phi'\}$.
- By the consequence rule, we can derive: $\vdash \{\phi\}\ s\ \{\phi'\}$:

$$\textsc{Consequence}\ \frac{\vdash \phi \Rightarrow \mathbf{wlp}(s, \phi') \qquad \vdash \{\mathbf{wlp}(s, \phi')\}\ s\ \{\phi'\} \qquad \vdash \phi' \Rightarrow \phi'}{\vdash \{\phi\}\ s\ \{\phi'\}}$$

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Dafny

- Dafny is a programming language and verification tool that allows developers to write code and specify its behavior using preconditions, postconditions, and invariants.
- It uses a combination of static verification and runtime checks to ensure that the code adheres to its specifications.
- Dafny supports features such as classes, methods, functions, and inductive datatypes, making it suitable for a wide range of applications.

Example – Factorial

```
method Factorial(n: nat) returns (x: nat)
  requires 0 <= n              // precondition
  ensures x == factorial(n)    // postcondition
{
  var i := n;
  x := 1;
  while 0 < i
    invariant 0 <= n  // invariants
    invariant 0 <= i <= n
    invariant x * factorial(i) == factorial(n)
  {
    x := x * i;
    i := i - 1;
  }
}
function factorial(n: nat): nat {
  if n == 0 then 1 else n * factorial(n - 1)
}
```

Example – Fibonacci

```
method ComputeFib(n: nat) returns (x: nat)
  requires 0 <= n          // precondition
  ensures x == fib(n)      // postcondition
{
  if n == 0 { return 0; }
  var i, a, b := 1, 0, 1;
  while i < n
    invariant 1 <= i <= n      // invariants
    invariant a == fib(i - 1)
    invariant b == fib(i)
  {
    a, b := b, a + b;
    i := i + 1;
  }
  return b;
}
function fib(n: nat): nat {
  if n == 0 then 0 else if n == 1 then 1 else fib(n - 1) + fib(n - 2)
}
```

Example – IndexOf

```
method IndexOf(a: array<int>, v: int) returns (index: int)
  ensures -1 <= index < a.Length
  ensures index != -1 ==> a[index] == v
  ensures index != -1 ==> forall k :: 0 <= k < index ==> a[k] != v
  ensures index == -1 ==> forall k :: 0 <= k < a.Length ==> a[k] != v
{
  var i := 0;
  while i < a.Length
    invariant 0 <= i <= a.Length
    invariant forall k :: 0 <= k < i ==> a[k] != v
  {
    if a[i] == v { return i; }
    i := i + 1;
  }
  return -1;
}
```

Next Lecture

- Separation Logic (1)

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